

Titanic dataset

Key stats:

-891 passengers

-38.4% survival rate

-65% male

-55% third class

-Southampton embarkation at 72.5%.

1. Gender was the strongest survival predictor: females survived at far higher rates than males across all age groups, consistent with the "women and children first" protocol.
 2. Passenger class strongly predicted survival: 1st class passengers had better outcomes than 3rd class
 3. Fare also affected survival rates with higher paying customers having a better chance at survival
 4. Small family groups (FamilySize 2–4) survived better than solo travellers or very large families. Passengers travelling alone had lower survival rates.
 5. Fare had 116 outliers (13%) and the highest right skew among numeric features
-

Housing dataset

Key stats: no missing values

- right-skewed price and area distributions

- 5 binary amenity features

-3-level furnishing status.

1. Area is the primary price driver, the scatter plot shows a clear positive trend.
2. Bathrooms show a stronger price relationship than bedrooms, price rises consistently with bathroom count, while bedrooms show more variance at higher counts.
3. Air conditioning and mainroad access likely produce the widest price gap.
4. Furnishing status creates a natural pricing tier (furnished > semi-furnished > unfurnished)